

Profile Summary

- **Software and Security Engineer** with **12** years of experience building **AI** serving infrastructure, sovereign cloud deployments, and identity data planes at **Google**, **OpenAI**, **AWS**, and **Twilio**.
- Specializes in **distributed systems reliability**, **AI model serving**, **identity and access management**, **incident response**, **DoS and denial-of-wallet mitigation**, and **ABAC**, using **load shedding**, **deadlines and retries**, **idempotency**, and **policy as code**.
- Broad toolkit covering languages (**C++**, **Python**, **Java**, **Lean 4**), cloud and containers (**Google Cloud**, **AWS**, **Kubernetes**, **Terraform**), and data stores (**Spanner**, **Postgres**, **DynamoDB**, **Aurora MySQL**).
- Strong **cross-functional** collaborator, partnering with **security**, **product**, and hundreds of **integrating service teams** to unblock launches and resolve issues under live incident pressure.
- Technical **leader** who has run teams of about **6** engineers, served as interim **manager**, and conducted around **100** interviews.

Technical Skills

Languages: C++, Python, Java, TypeScript, JavaScript, SQL, Lean 4, Apex

Distributed Systems: load shedding, deadlines and retries, idempotency, load testing, A/B testing

Messaging & Observability: Kafka, Amazon SQS, Google Cloud Pub/Sub, Datadog, Sentry, black-box monitoring (probers)

Reliability: incident response, postmortems, dashboards and alerting, profiling, distributed tracing

Security: AWS IAM, ABAC, just-in-time access, policy as code, threat modeling, C++ memory safety, privacy, data residency

Security & Compliance: denial-of-service and denial-of-wallet mitigation, insider-risk access controls, red teaming, PCI DSS, sanctions screening, AML risk scoring, GRC compliance automation

AI Infrastructure: Gemini and Vertex AI model serving, vLLM, AI accelerator capacity, Amazon Bedrock

AI Tooling: LLM-assisted code analysis, Claude Code

Formal Verification: Lean 4, machine-checked proofs, LLM-assisted proof development

Cloud & Infrastructure: Google Cloud (GKE, Vertex AI), AWS (DynamoDB, SQS, Lambda, Aurora MySQL, CloudFormation, Glue), Azure

Containers & APIs: Kubernetes, Docker, Terraform, GitHub Actions, FastAPI, REST APIs, gRPC

Frameworks, Storage & Data: Salesforce (Lightning, SOQL), Spanner, Postgres, MySQL, DynamoDB, Redis, Databricks, ETL pipelines

Work Experience

Google Software Engineer

Sunnyvale, CA • Jul. 2025 - Present

- Protected **distributed-systems reliability**, **security**, and **privacy** across **Gemini** and **Vertex AI** serving infrastructure, securing **C++** serving paths against abuse and outages and extending model serving into an isolated cloud, improving trust in **Google's AI** platforms.
- Collaborated across **Cloud Security**, **Billing**, **Product**, **Privacy**, and **Legal** on design reviews, launch readiness, and oncall handoffs, while leading **code reviews**, mentoring around **10** engineers, onboarding new **TLS/leaders**, and presenting findings to **engineering peers** and **senior stakeholders** to align priorities and secure approvals.

Gemini serving data plane: distributed systems reliability and security (Apr 2026 - Present)

- Hardened the **C++** serving path against **denial-of-service** and **denial-of-wallet** cost exhaustion through deadline propagation, early rejection of expired requests, adaptive load shedding, and consumption caps, keeping the service available during overload.
- Diagnosed an **AI** accelerator defect via trace analysis, utilization correlation, and metric discrepancies, patched it with **HTTP cancellation propagation**, and drove the incident and postmortem with flag-flip mitigation and extensive safety testing.
- Moved retries onto a standard framework with exponential backoff, jitter, and service-wide retry budgets through a major refactor, then **A/B** load-tested identical synthetic traffic across replicas without customer data.
- Shipped the production fix for an out-of-memory incident using heap profiling, core dumps, and agentic analysis, adding memory-efficiency improvements, load shedding, observability, task scaling, and a crash-to-root-cause dashboard.
- Delivered data residency through region-pinned request tagging and location-aware backend selection, plus privacy and memory safety through **absl::Span** migration and **ASan/MSan** CI gating.

Vertex AI on Google Cloud Dedicated, an isolated sovereign cloud (Jul 2025 - Mar 2026)

- Designed the infrastructure plan for **Vertex AI Model Garden** serving in **Google Cloud Dedicated**, a fully isolated sovereign cloud with no public **Google Cloud** dependency, and drove its first test control-plane deployment by establishing **IAM**, **ACL**, and turn-up procedures where no runbook existed.
- Mapped service dependencies through build-graph traversal and service-interface enumeration using agentic **Gemini** analysis, structured **JSON**, parallel package fan-out, and **Python** scripts, producing a **100+** task backlog executed in parallel.
- Built the team's first black-box monitoring using golden-path synthetic probes against per-region endpoints and probe-derived availability signals for future **SLOs**, setting the standard for **Vertex** services in that cloud.
- Enforced dependency-injection standards through custom reflection-based presubmit tests and drove **AI** coding tool adoption through setup guides and custom agents.

OpenAI Member of Technical Staff

San Francisco, CA • Sep. 2024 - Apr. 2025

- Stepped into a reliability emergency on the internal just-in-time access service used by nearly all employees, diagnosing latency spikes and unattributed failures and hardening monitoring via **Datadog APM tracing**, **RED metrics**, and data-health validation pipelines, cutting **p99 latency by over an order of magnitude**.
- Instrumented **p99 latency** and error rates in **Datadog** with custom metrics, profiling, tracing, and composite monitors, establishing the monitoring baseline for security services and cutting recovery time from hours to minutes in one case.
- Eliminated duplicate records and constraint violations through idempotency keys, **ON CONFLICT** upserts, **SELECT FOR UPDATE** locks, and isolation-level tuning, while maintaining the policy-as-code pipeline on a custom **Terraform** provider and **GitHub Actions** and co-authoring the postmortem for a policy-migration incident.
- Automated inactive service-account detection with **Databricks** and **PySpark** by joining **Entra ID** sign-in logs and **IAM** credential-last-used reports, served on the company-wide security on-call rotation, and enforced least-privilege approvals through **Terraform** access checks.

Amazon Web Services

Senior Software Development Engineer, AWS Identity (IAM control plane)

Denver, CO • Nov. 2022 - Aug. 2024

- Directed **6** engineers as tech lead of an **IAM** control plane authorizing every **AWS API** call at hundreds of millions per second per **AWS** public figures, owning policy-evaluation **RFCs**, caching design, **p99.9** latency targets, and availability budgets.
- Treated availability as a security property, automating issue resolution through auto-rollback and alerting, migrating configuration to **CDK** and **CloudFormation**, and adding monitoring and tests across every **AWS** region, cutting the team's on-call load.
- Drove **attribute-based access control (ABAC)** strategy across organizations, principals, tags, and tag policies for hundreds of **AWS** service teams integrating with **IAM**, repairing broken authorization-stack integrations via static code analysis.
- Found zero-day vulnerabilities through **red teaming**, mitigating them through patches and re-architecture, and in **2023** applied **Claude on Amazon Bedrock** to catch integration errors missed by static analysis.

Twilio Staff Software Engineer

Denver, CO • Jul. 2021 - Oct. 2022

- Managed **6** engineers as tech lead and **4** months as interim **engineering** manager on the billing platform, migrating it to **Amazon Aurora MySQL** while processing over **1** billion daily billing transactions.
- Diagnosed defects in a half-finished **Kafka-to-SQS** migration at maximum utilization, using queueing theory to predict throughput limits, win prioritization for the fix, and avert month-end close delays and message loss.
- Owned the **ETL** workstream for revenue data on **AWS Glue** with **Spark**, guaranteeing correctness through reconciliation jobs and working with security to approve a technology new to **Twilio**.
- Hardened a legacy billing service through query and pagination rebuilds, **UTC** normalization and spring-forward regression tests, lower log costs, and restricted billing-data access under **PCI DSS**.

Amazon Software Development Engineer

Denver, CO • Jul. 2019 - Jul. 2021

- Shipped retail web service changes to hundreds of millions of customers on a legacy stack with limited **rollback**, spanning order and fulfillment **pipelines**, frontend, and backend to enable ordering experiences like digital items linked to physical items returned together.

Google Software Engineer

Boulder, CO • Dec. 2015 - Jun. 2019

- Migrated **sanctions screening** from the legacy **Google Payments** case-review platform over **6** months and **9,000+** lines of code, automating document **requests** and cutting manual first-level operations reviews **61%**.
- Built a prototype **migration tool** and co-wrote a framework guide used by **50** teams, advanced an **A/B testing** framework for Payments latency regressions, and led a tool reviewing **26,000** Street View images weekly, cutting page latency from **8-9** to **1-2** seconds.
- Designed a nightly batch service ranking Payments customers into **5** monthly-volume tiers, live since **2016** and still in production in **2026**, and completed **65** **Java** readability **code reviews** in **6** months as a certified reviewer.

Trifecta Technologies Software Developer

Allentown, PA • Dec. 2014 - Dec. 2015

- Developed **Salesforce e-commerce** and booking systems for an events client using **Apex**, **SOQL**, and **JavaScript**.

Projects

Overload

Jul. 2026

- Built an **Apache 2.0** open-source **Lean 4** library with **700** verified theorems and definitions on retry-driven congestion collapse and safe retry limits, using **Claude Code** while owning modeling and verification, with the paper in **Google's** publication review.

Education & Publications

Muhlenberg College B. Sc. in Computer Science

Allentown, PA • Aug. 2010 - May 2014

Evaluating Software Safety Standards: A Systematic Review and Comparison

2014